Integração em Sistemas Inteligentes
Exam Model (Version 1) Number: _____
2021-02-01
Duration: 2 Hours Name: _____

This exam contains 4 pages and 31 questions.
Students must put their number and name at the beginning of the exam and on the draft sheets provided to them.
- The use of any type of consultation is not allowed.
- All material provided by the teacher must be delivered at the end of the test
- In multiple choice questions, only one is correct.
- To answer multiple-choice questions, you should mark your answer with a circle over the desired paragraph.
- Note that each incorrect answer **discounts** $\frac{1}{4}$ from your quote on multiple choice questions.
- In order to answer the True / False questions, you must check your answer T or T over the lines.
- In the end, you must give the teacher all the proof and draft sheets.

1. (2%) Which of the following activities is typical in an Investigation Process?
   A. Academic Writing;
   B. Communication;
   C. Conferences;
   D. All of them.
2. (2%) The IMRaD approach on writing a scientific research report is a: A. Short written text designed for publication in scientific journal reporting work in progress; B. Paper that is structured by four main sections: Introduction, Methods, Results, and Discussion; C. Document that summarizes the main findings and comments on those findings; D. Paper that is structured by four main sections: Introduction, Materials, Research, and Discussion.
3. (2%) The abstract section in a paper is a: A. Formal lesson or talk in which a teacher or expert addresses a (passive) audience; B. Long text produced by postgraduate students to achieve a higher degree; C. Short version of a longer text (informal or formal/oral or written). D. None of them.
4. (2%) The statement "explains why this research is important or necessary or important"is concerned with the section: A. Introduction; B. Methods; C. Results; D. Conclusion.
5. (2%) The right statement about Cyber Physical Systems is: A. Results from the integration of computation and physical processes; B. Computer system in which a mechanism is controlled or monitored by computer-based algorithms; C. Physical and software components are deeply intertwined; D. All of them.
6. (2%) Examples of Hadoop wrappers includes: A. Hive; B. Apache Pig; C. None of them; D. All of them.
7. (2%) The right statement about Supervisory Control and Data Acquisition (SCADA) is:
   A. Used to automate and control industrial processes;
   B. Includes Power plants or assembly lines;
   C. Includes power generation and distribution, gas and oil pipelines, and water and waste management;
   D. All of them.
8. (2%) SCADA components does not includes: A. PLCs; B. RTUs; C. Historian Database servers (HDB); D. Gateways.
9. (2%) Hundreds to thousands of “sensor nodes” communicating with each other constitutes a:
   A. ZigBee Alliance;
   B. Wireless Sensor Networks (WSNs);
   C. Internet of Things (IoT) cluster;
   D. None of them.
10. (2%) The correct statement on Remote Terminal Unit (RTU) is: A. Microprocessor-controlled electronic device; B. Interfaces objects in the physical world to a distributed control system or SCADA system; C. Deployed for wide geographic telemetry; D. All of them.
11. (2%) In Sensor Operating Systems:
    A. Event driven and Multi threaded are Design concepts;

B. TinyOS, Contiki and RIOT are OSs examples;

C. All of them;

D. None of them.

12. (2%) An example of Malware is:
    A. Stuxnet;
    B. BlackEnergy;
    C. None of them;
    D. All of them;
13. (2%) Defensive actions in SCADA does not include:
    A. Intrusion and Anomaly Detection System (IADS);
    B. Intrusion detection system (IDS);
    C. Reverse Proxy;
    D. All of them.
14. (2%) The correct statement on Network Intrusion Detection Systems (NIDS)? A. Includes monolithic applications where modules are highly coupled; B. It can receive and publish messages from the IoT devices; C. Snort, Suricata or Bro are examples; D. All of them.
15. (2%) Critical Infrastructures (CI)s Testbeds are important for: A. Vulnerability assessment ; B. Impact analysis; C. Attack-defense evaluation and validation; D. All of them.
16. (2%) The purpose of Smart Grids is: A. To reduce energy consumption and costs; B. Increase reliability and transparency; C. Enable data acquisition and remote control between control centers and a large number of power grid facilities; D. All of them.
17. (2%) Machine Learning (ML) is a: A. Function that has the input-output behavior observed in the data; B. Function Spcefically applied to Neurobiology; C. A set of algorithms; D. Computer program that automatically improves, supported by the insights retrieved from the experience.
18. (2%) The incorrect statement is:
    A. Computational theory is not used in ML;
    B. ML is one of the branches in Artificial intelligence;
    C. Information theory is used in ML;
    D. All of them.
19. (2%) Classification is: A. A learning model that summarizes data with a set of parameters of fixed size (independent of the number of training examples); B. the training data to estimate a fixed set of parameters; C. the predicted outcome; D. All of them.
20. (2%) MapReduce is a processing technique for: A. reading data from HDFS, process and generates some intermediate results; B. including two important stages, Map and Reduce; C. summarizing data with a set of parameters of fixed size; D. All of them.
21. (2%) In Unsupervised Learning:
    A. Example labeled inputs are provided with with their desired outputs;
    B. Example inputs are not provided;
    C. Is the approach to consider performance indicators;
    D. All of them are correct.
22. (2%) A test result incorrectly indicates a disease when the disease is not present, is a:
    A. False Positive;
    B. True Positive;
    C. False Negative;
    D. True Negative.
23. (2%) If a Model does not generalise well to make predictions on new and unseen is: A. Overriding: B. Overfitting; C. Regularization; D. None of them.
24. (2%) The purpose of the objective function is: A. to evaluate a candidate solution; B. to evaluate the cost function; C. find the labels; D. All of them.
25. (2%) Application programming interface (API):
    A. is as abstraction layer used to connect code components;
    B. is as abstraction layer used to connect code components in the same programming language;

C. is an interface to plug executables;

D. None of them.

26. (9%) Discuss the common IoT Architectures layers.

As arquiteturas IoT mais comuns têm 3 a 5 camadas:

Perceção / Dispositivos: sensores e atuadores que recolhem dados do mundo físico e atuam sobre ele (ex.: RFID, temperatura, nós WSN).

Rede / Transporte: transmite os dados dos dispositivos até à plataforma, através de ZigBee, Wi-Fi, Bluetooth, 6LoWPAN, LoRa, redes celulares ou gateways. Usa protocolos como MQTT e CoAP.

Middleware / Processamento: armazena, filtra e processa os dados (cloud/edge, bases de dados, streaming, análise e ML). Abstrai o hardware e faz a comunicação entre componentes.

Aplicação: serviços ao utilizador final, como smart grids, smart cities, saúde e indústria.
Negócio: gestão global do sistema, dashboards, modelos de negócio, segurança e privacidade.

A segurança é transversal a todas as camadas.

27. (13%) Evaluate each of the following statements, **indicating whether it is True (T) or False (F)**:

(a) F Just a minor volume of "Big data" is unstructured.

(b) F Constrained Application Protocol (CAP) is a specialized web transfer protocol for use with constrained nodes and constrained networks.

(c) F The same programming languages can be used in the implementation of API data integration models.

(d) F Kafka is a Machine Learning Model.

(e) T Apache Hadoop and Apache Spark are examples of Horizontal Scaling Platforms.

(f) T MapReduce breaks the entire scheme into mappers and reducers.

(g) F Volume and Velocity are the only Big Data characteristics.

(h) F Scaling is the system integration ability.

(i) F A NoSQL database Data is available in a relational format.

(j) T Hadoop relies on an application layer to deliver high-availability to detect and handle failures.

(k) T A Proxy is an intermediary that mainly is concerned with forwarding requests and relaying back responses.

(l) F A Cross-Proxy is a proxy that translates between different protocols.

(m) T Reverse-Proxy is an endpoint that stands in for one or more other servers and satisfies requests on behalf of these, doing any necessary translations.

(n) F Human intervention is required in Machine to Machine (M2M).

(o) F NoSQL databases rely on schemas.

(p) F Log data lacks any proper structure in their events.

(q) T Hadoop is a batch event storage and processing system.

(r) F Installing more processors, more memory, and faster hardware typically within a single server is an example of horizontal scaling.

(s) F Vertical scaling involves distributing the workload across many servers.

(t) T Streaming and Queuing is due with continuously massively and at great speed produced data by multiple sources and then routed to receivers.

(u) T Message Passing Interface (MPI) allows the implementation of decentralized distributed network architecture.

(v) F Resources, memory, battery, processing power, range, are examples of ZigBee protocol challenges.

(w) F Contiki-NG is an Architectural Framework for IoT.

(x) T HDFS (Hadoop Distributed File System) allows storing data across a cluster of commodity machines while providing high availability and fault tolerance.

(y) T Kafka, ZooKeeper and RabbitMQ are examples of Streaming and Queuing technologies.

(z) T One horizontal scaling drawback is due to the software that has to handle all the data distribution and parallel processing complexities.

28. (11%) Evaluate each of the following statements, **indicating whether it is True (T) or False (F)**

(a) F In Message Queuing Frameworks communication protocol can be decoupled by using queues containing APIs.

(b) T A Message Queuing Framework broker can perform load balancing and job distribution.

(c) F IEEE P2413 is an Open-source, cross-platform operating system for Next-Generation IoT devices.

(d) T Distributed message queuing frameworks take care of low-level distributed system management complexities.

(e) F MapReduce is a data processing scheme used in Hadoop involving partition and combine stages.

(f) ___T___ Spark represents the next generation paradigm for big data processing as an alternative to Hadoop.

(g) ___T___ Middleware layer maintains the communication between the different system components.

(h) ___T___ A modern requirement in data integration is to overpower the limits of point-to-point communication in a nonsynchronous fashion.

(i) ___T___ Message queuing is an architectural pattern allowing time-space-and synchronization-decoupling.

(j) ___T___ Hadoop YARN is a resource management layer scheduling jobs across the cluster.

(k) ___F___ Event Streaming Platform acts as a proxy for data streams.

(l) ___F___ MapReduce Mappers aggregate the intermediate results to generate the final output and write it to the HDFS.

(m) ___T___ A Event Streaming Platform acts as a buffer between systems.

(n) ___T___ Stream processing enables continuous, real-time applications built to react to, process, or transform streams.

(o) ___F___ Apache Pig is a popular MapReduce wrapper but not Hive.

(p) ___T___ The event streaming platform captures streams of events or data changes and feeds these to other data systems such as relational databases, key-value stores, Hadoop, or the data warehouse.

(q) ___T___ OpenAPI defines a standard, language-agnostic interface to RESTful APIs.

(r) ___T___ In the publisher-subscriber approach, consumers of data can come and go and are fully decoupled from the source.

(s) ___T___ Cooja is a network simulator specifically designed for Wireless Sensor Networks.

(t) ___T___ KAFKA is a distributed publish-subscribe messaging system that maintains topics in partitioned and replicated feeds of messages.

29. (5%) Discuss Vertical and Horizontal scaling.

Vertical (scale-up): aumenta a capacidade de um único servidor, com mais CPU, memória ou discos. É simples de implementar e não exige alterações ao software. Tem um limite físico, o custo cresce depressa e o servidor é um ponto único de falha.
Horizontal (scale-out): distribui a carga por muitos servidores (cluster de máquinas comuns), como fazem o Hadoop e o Spark. Tem escalabilidade quase ilimitada, custo mais baixo e tolerância a falhas. O inconveniente é a complexidade do software, que tem de tratar da distribuição dos dados e do processamento paralelo.

30. (6%) Distinguish between Supervised and Unsupervised Learning.

Supervisionada: os dados de treino incluem exemplos com as saídas desejadas (labels). O modelo aprende um mapeamento entrada→saída para prever em dados novos. Os problemas típicos são classificação e regressão (ex.: SVM, regressão logística, árvores de decisão). Avalia-se com métricas como exatidão, precisão e recall.
Não supervisionada: os dados não têm labels. O objetivo é descobrir estrutura ou padrões escondidos, como grupos, associações ou redução de dimensionalidade (ex.: k-means, PCA, deteção de anomalias).

A diferença essencial é a existência ou não de saídas conhecidas no treino. A aprendizagem supervisionada exige dados rotulados, que são caros de obter, enquanto a não supervisionada é mais exploratória e a sua avaliação é mais subjetiva.

31. (6%) Discuss Microservices in Data Integration.

Os microserviços dividem uma aplicação em serviços pequenos, independentes e com uma responsabilidade bem definida. Cada um pode ser desenvolvido, implantado e escalado separadamente, em qualquer linguagem. Comunicam por APIs (REST/OpenAPI) ou por mensagens (filas, Kafka, publish-subscribe).

Na integração de dados, cada microserviço pode tratar de uma tarefa, como ingestão, transformação, validação ou ligação a uma fonte. Isto traz:

baixo acoplamento e comunicação assíncrona, superando os limites do ponto-a-ponto;
escalabilidade horizontal por serviço e tolerância a falhas isoladas;
flexibilidade tecnológica e implantação contínua (contentores, orquestração).

Em contrapartida, aumenta a complexidade operacional, de monitorização e de consistência de dados, em comparação com aplicações monolíticas fortemente acopladas.

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	Total
Quotation	2	2	2	2	2	2	2	2	2	2	2	2	2	26
Evaluation														

Question	14	15	16	17	18	19	20	21	22	23	24	25	Total
Quotation	2	2	2	2	2	2	2	2	2	2	2	2	24
Evaluation													

Question	26	27	28	29	30	31	Total
Quotation	9	13	11	5	6	6	50
Evaluation							